

Contents

1	More Linear Circuit elements	3
1.1	Capacitor	4
1.2	Inductor	9
1.3	Operational amplifier	14
1.3.1	Introduction	14
1.3.2	Ideal Op-amp	19
1.4	Practical Op-amp circuits	21
1.4.1	Basic mathematical operations with op-amp	21
1.4.2	Difference Amplifier	29
1.4.3	Cascading op-amp circuits	34
1.4.4	Op-amp circuits with capacitors	37

Chapter 1

More Linear Circuit elements

TABLE 6.1

Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{\text{eq}} = R_1 + R_2$	$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{\text{eq}} = L_1 + L_2$
Parallel:	$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{\text{eq}} = C_1 + C_2$	$L_{\text{eq}} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

[†]Passive sign convention is assumed.

1.1 Capacitor

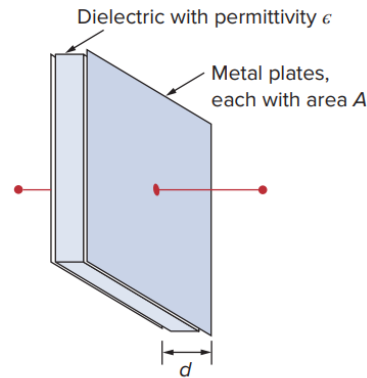


Figure 6.1
A typical capacitor.

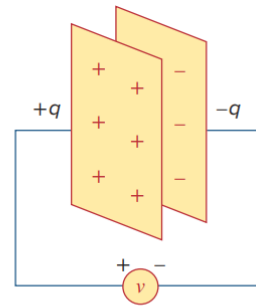


Figure 6.2
A capacitor with applied voltage v .

1. A **capacitor** is a passive element designed to **store energy** in its electric field. (Fig. 6.1) A capacitor consists of two conducting plates separated by an insulator (dielectric). The conducting plates may be aluminum foil, while the dielectric may be air, ceramic, paper, or mica.
2. (Fig. 6.2) The capacitor stores the electric charge q whose amount is directly proportional to the applied voltage v .

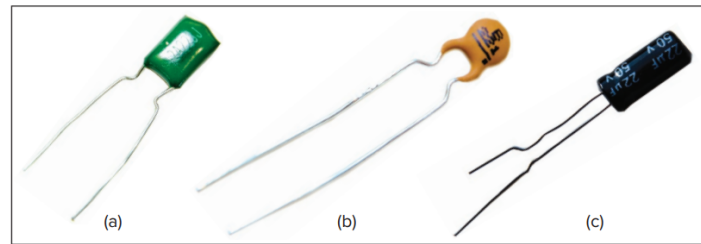
$$q = Cv$$

Capacitance C is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F). ($1 \text{ F} = 1 \text{ coulomb/volt}$). Capacitance does not depend on q or v . It depends on the physical dimension of the capacitor:

$$C = \epsilon \frac{A}{d}$$

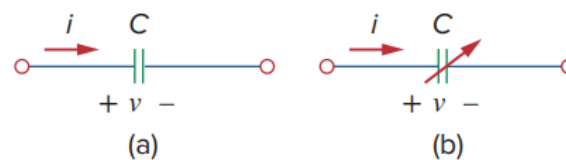
- The surface area of the plates—the larger the area, the greater the capacitance.
- The spacing between the plates—the smaller the spacing, the greater the capacitance.
- The permittivity of the material—the higher the permittivity, the greater the capacitance

Typical capacitance values range from picofard (pF) to microfarad (μF).

**Figure 6.4**

Fixed capacitors: (a) polyester capacitor, (b) ceramic capacitor, (c) electrolytic capacitor.
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3. (Fig. 6.4) The common types of fixed-value capacitors include polyester capacitors, which are light-weight, stable, and predictable change with temperature. Film capacitors are rolled and housed in metal or plastic films. Electrolytic capacitors produce very high capacitance.

**Figure 6.3**

Circuit symbols for capacitors: (a) fixed capacitor, (b) variable capacitor.

4. **Voltage polarity of the capacitor.** By passive sign convention, positive current flows toward the (+) (positive) plate, thereby charging the capacitor:
- If $v > 0$, $i > 0$ or $v < 0$, $i < 0$, the capacitor is charging. Charges accumulate on each plate until the voltage across the capacitor balances the external DC voltage, at which point the current ceases.
 - If $v \cdot i < 0$, the capacitor is discharging. The current reduces the accumulated charge on each plate, eventually restoring electrical neutrality.
5. A capacitor satisfies the following **current-voltage relationship** (assuming the passive sign convention),

$$i = \frac{dq}{dt} = C \frac{dv}{dt}$$

If the capacitance is independent of voltage, this capacitor is said to be linear. Conversely, the voltage-current relationship can be obtained by integration,

$$v = \frac{1}{C} \int_{-\infty}^t i \, dt = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$$

According to this relation, the capacitor has **memory**: the capacitor voltage depends on the **past history** of the capacitor current.

6. The instantaneous power delivered to the capacitor is

$$p = vi = v \left(C \frac{dv}{dt} \right) = Cv \frac{dv}{dt}$$

Hence, the energy stored in the capacitor is therefore

$$W = \int_{-\infty}^t p \, dt = \int_{-\infty}^t C v \frac{dv}{dt} dt = \int_{-\infty}^t C v \, dv = \frac{1}{2} C [v(t)^2 - v(-\infty)^2]$$

If the capacitor is initially uncharged (or fully discharged), $v(-\infty) = 0$ and

$$W = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

This represents the energy stored in the electric field that exists between the plates of the capacitor. This energy can be **retrieved** since an ideal capacitor cannot dissipate energy. The word *capacitor* is derived from this element's capacity to store energy in an electric field.

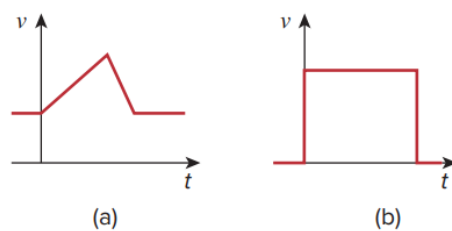


Figure 6.7

Voltage across a capacitor: (a) allowed, (b) not allowable; an abrupt change is not possible.

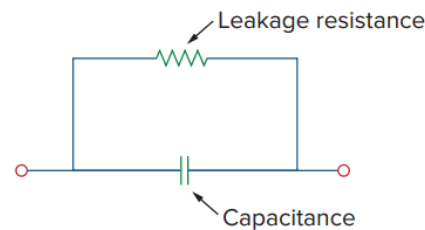


Figure 6.8

Circuit model of a nonideal capacitor.

7. The **current-voltage relationship** of the capacitor captures key behaviors that are essential for analyzing circuits involving capacitive elements.

- A capacitor is an **open** circuit to DC. Under DC condition, and

$$\frac{dv}{dt} = 0 \text{ (DC)} \rightarrow \therefore i = C \frac{dv}{dt} = 0$$

- The voltage on the capacitor cannot change abruptly. The capacitor resists an abrupt change in the voltage across it. One mathematical way to see this property is

$$\frac{dv}{dt} \rightarrow \infty \rightarrow i = C \frac{dv}{dt} \rightarrow \infty$$

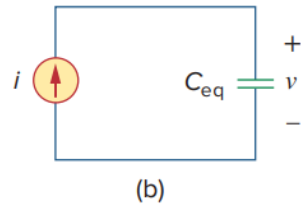
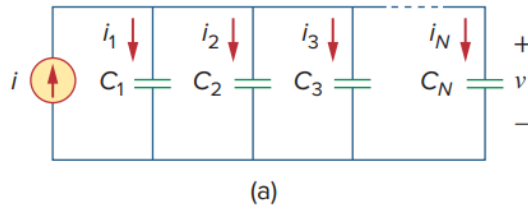
However, the time derivative of voltage (the abrupt change of voltage across capacitor) cannot become infinity because infinite current is physically impossible. Fig. 6.7 illustrate two scenarios of the voltage across a capacitor that can be allowed or not allowable.

An alternative way of looking at this is using the fact that the work done to charge a capacitor depends quadratically on the voltage across it,

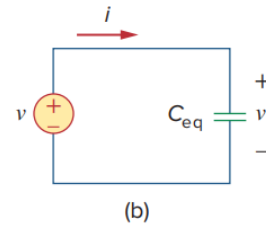
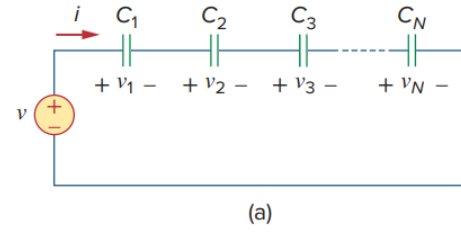
$$W = \frac{1}{2} C v^2$$

Since injecting or extracting energy can only be done over some finite time, voltage cannot change instantaneously across a capacitor.

8. Series and parallel capacitors

**Figure 6.14**

(a) Parallel-connected N capacitors,
(b) equivalent circuit for the parallel capacitors.

**Figure 6.15**

(a) Series-connected N capacitors,
(b) equivalent circuit for the series capacitor.

- (Fig. 6.14) Parallel N capacitors have the **same voltage** v across them. Applying KCL in Fig. 6.14a,

$$\begin{aligned}
 i &= i_1 + i_2 + \cdots + i_N \\
 &= C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt} \\
 &= (C_1 + C_2 + \cdots + C_N) \frac{dv}{dt} \\
 &= C_{eq} \frac{dv}{dt} \\
 \therefore C_{eq} &= C_1 + C_2 + \cdots + C_N
 \end{aligned}$$

- (Fig. 6.15) Series N capacitors have the **same current** i through them. Applying KVL in Fig. 6.15a,

$$\begin{aligned}
 v &= v_1 + v_2 + \cdots + v_N \\
 &= \left[\frac{1}{C_1} \int_{t_0}^t i \, dt + v_1(t_0) \right] + \left[\frac{1}{C_2} \int_{t_0}^t i \, dt + v_2(t_0) \right] + \cdots + \left[\frac{1}{C_N} \int_{t_0}^t i \, dt + v_N(t_0) \right] \\
 &= \left[\left(\frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_N} \right) \int_{t_0}^t i \, dt + (v_1(t_0) + v_2(t_0) + \cdots + v_N(t_0)) \right] \\
 &= \frac{1}{C_{eq}} \int_{t_0}^t i \, dt + v(t_0) \\
 \therefore \frac{1}{C_{eq}} &= \left(\frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_N} \right)
 \end{aligned}$$

9. **Example.** For the circuit in Fig. 6.18, find the voltage across each capacitor.

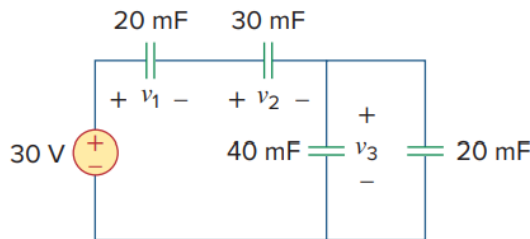


Figure 6.18
For Example 6.7.

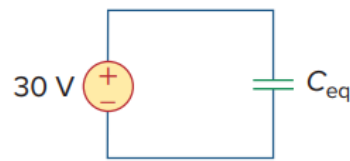


Figure 6.19
Equivalent circuit for Fig. 6.18.

Solution:

We first find the equivalent capacitance C_{eq} , shown in Fig. 6.19. The two parallel capacitors in Fig. 6.18 can be combined to get $40 + 20 = 60$ mF. This 60-mF capacitor is in series with the 20-mF and 30-mF capacitors. Thus,

$$C_{eq} = \left(\frac{1}{60} + \frac{1}{30} + \frac{1}{20} \right)^{-1} \text{ mF} = 10 \text{ mF}$$

The total charge is

$$q = C_{eq}v = 10 \times 10^{-3} \times 30 = 0.3 \text{ C}$$

This is the charge on the 20-mF and 30-mF capacitors, because they are in series with the 30-V source. (A crude way to see this is to imagine that charge acts like current, since $i = dq/dt$.) Therefore,

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V} \quad v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

Having determined v_1 and v_2 , we now use KVL to determine v_3 by

$$v_3 = 30 - v_1 - v_2 = 5 \text{ V}$$

Alternatively, since the 40-mF and 20-mF capacitors are in parallel, they have the same voltage v_3 and their combined capacitance is $40 + 20 = 60$ mF. This combined capacitance is in series with the 20-mF and 30-mF capacitors and consequently has the same charge on it. Hence,

$$v_3 = \frac{q}{60 \text{ mF}} = \frac{0.3}{60 \times 10^{-3}} = 5 \text{ V}$$

1.2 Inductor

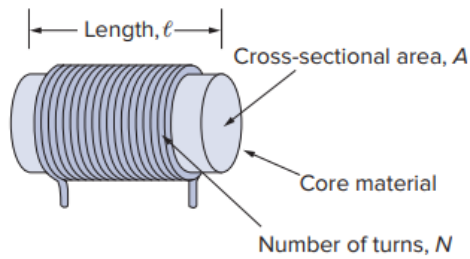


Figure 6.21
Typical form of an inductor.

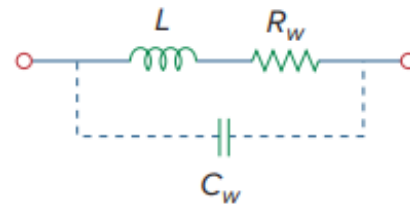


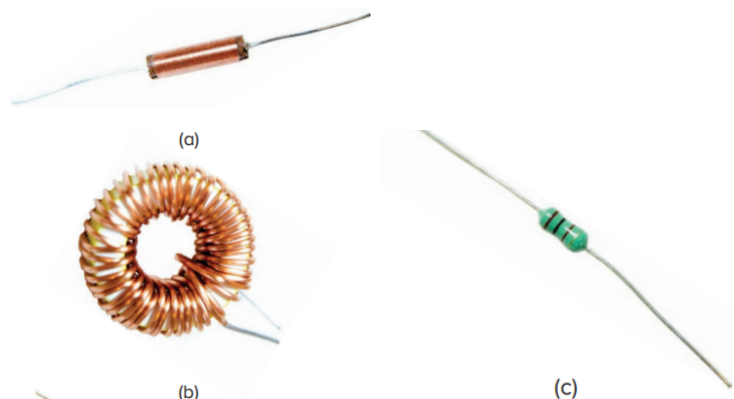
Figure 6.26
Circuit model for a practical inductor.

1. A **inductor** is a passive element designed to **store** energy in its **magnetic field**.
2. Any conductor of electric current has inductive properties. But in order to enhance the inductive effect, a practical inductor is usually formed into a cylindrical coil with many turns with conducting wire (Fig. 6.21).
In the nonideal inductor, the winding resistance R_w and the winding capacitance C_w (due to capacitive coupling between the conducting coils) are present. C_w can be ignored in most cases except at high frequencies.
3. **Inductance** is the property whereby an inductor exhibits **opposition** to the change of current flowing through it, measured in henrys (H). ($1H = 1 \text{ volt sec per ampere}$). The inductance depends on its physical dimension and construction,

$$L = \mu N^2 \frac{A}{l}$$

(N : the number of turns, l is the length, A is the cross-sectional area, μ is the permeability of the core. See Fig. 6.21). If the inductance does not depend on the current, the inductor is said to be **linear**.

4. Typical practical inductors have inductance values ranging from a few microhenrys (μH) in communication systems, to tens of henrys as in power systems. Various types of inductors are shown below: (a) solenoidal wound inductor, (b) toroidal inductor, (c) axial lead inductor.



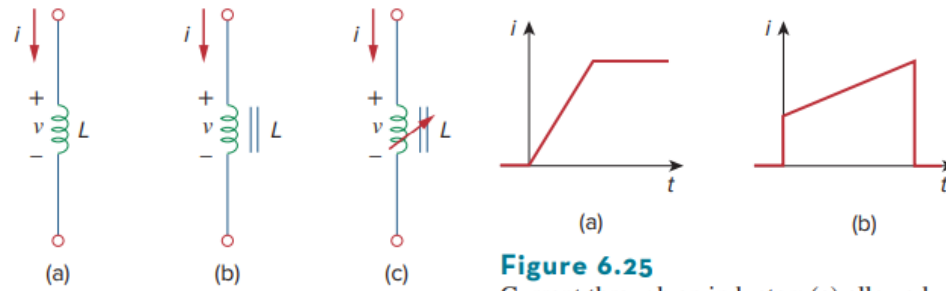


Figure 6.23
Circuit symbols for inductors: (a) air-core,
(b) iron-core, (c) variable iron-core.

Figure 6.25
Current through an inductor: (a) allowed,
(b) not allowable; an abrupt change is not possible.

5. **Voltage-current relationship.** If the current is allowed to pass through an inductor, the voltage across inductor is given by

$$v = L \frac{di}{dt}$$

Note the voltage polarity of the inductor in Fig. 6.23. The electromotive force(induced electric potential) is developed to resist the change of the current flowing it. This relationship of the inductor captures key behaviors that are essential for analyzing circuits involving inductive elements:

- An inductor acts like a **short** circuit to DC.

$$\frac{dI}{dt} = 0 \text{ (DC)} \rightarrow v = L \frac{dI}{dt} = 0$$

- The current through an inductor cannot change abruptly, since a discontinuous change of current through an inductor requires an infinite voltage, which is not physically possible.

$$\text{if } \frac{dI}{dt} \rightarrow \infty, v = L \frac{dI}{dt} \rightarrow \infty$$

However, the voltage over inductor can change abruptly.

6. **Current-voltage relationship.**

$$di = \frac{v}{L} dt \rightarrow i = \frac{1}{L} \int_{-\infty}^t v dt = \frac{1}{L} \int_{t_0}^t v dt + i(t_0)$$

7. **Energy stored in an inductor.** The instantaneous power delivered to an inductor, which stores energy in its magnetic field, is

$$p = vi = \left(L \frac{di}{dt} \right) i$$

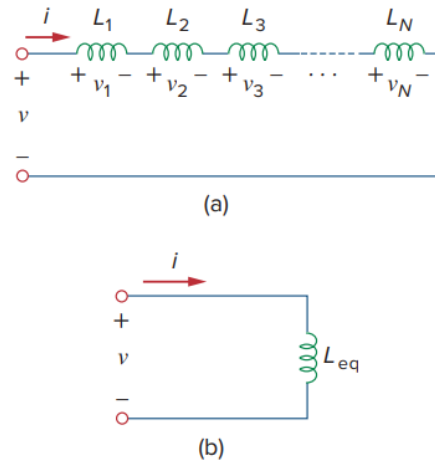
Then the energy stored is

$$W = \int_{-\infty}^t p dt = \int_{-\infty}^t Li \frac{di}{dt} dt = \frac{1}{2} L [i(t)^2 - i(-\infty)^2]$$

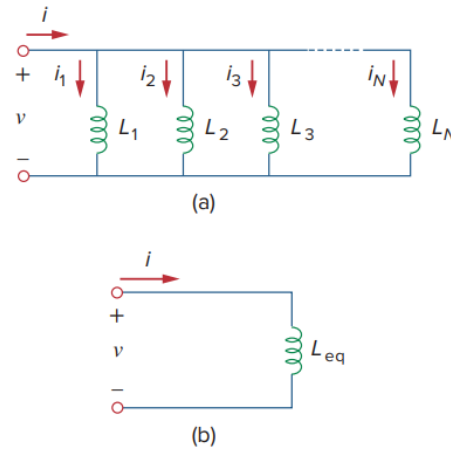
If $i(-\infty) = 0$,

$$W = \frac{1}{2} Li^2$$

8. Series and parallel inductors

**Figure 6.29**

(a) A series connection of N inductors,
(b) equivalent circuit for the series inductors.

**Figure 6.30**

(a) A parallel connection of N inductors,
(b) equivalent circuit for the parallel inductors.

- Series N inductors share **the same current**. By KVL,

$$\begin{aligned}
 v &= v_1 + v_2 + \cdots + v_N \\
 &= L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + \cdots + L_N \frac{di}{dt} \\
 &= (L_1 + L_2 + \cdots + L_N) \frac{di}{dt} \\
 &= L_{eq} \frac{di}{dt} \\
 \therefore L_{eq} &= L_1 + L_2 + \cdots + L_N
 \end{aligned}$$

- Parallel N inductors share **the same voltage**. By KCL,

$$\begin{aligned}
 i &= i_1 + i_2 + \cdots + i_N \\
 &= \left[\frac{1}{L_1} \int_{t_0}^t v \, dt + i_1(t_0) \right] + \left[\frac{1}{L_2} \int_{t_0}^t v \, dt + i_2(t_0) \right] + \cdots + \left[\frac{1}{L_N} \int_{t_0}^t v \, dt + i_N(t_0) \right] \\
 &= \left[\left(\frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_N} \right) \int_{t_0}^t v \, dt + (i_1(t_0) + i_2(t_0) + \cdots + i_N(t_0)) \right] \\
 &= \frac{1}{L_{eq}} \int_{t_0}^t v \, dt + i(t_0) \\
 \therefore \frac{1}{L_{eq}} &= \left(\frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_N} \right)
 \end{aligned}$$

9. Example. Computing stored energies in DC condition. Solution:

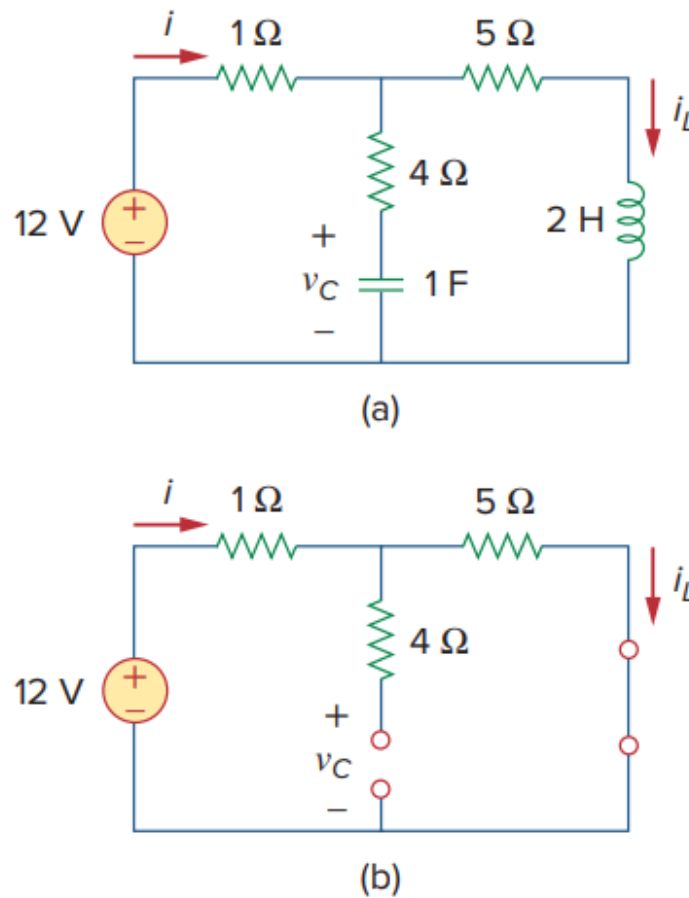


Figure 6.27
For Example 6.10.

- (a) Under dc conditions, we replace the capacitor with an open circuit and the inductor with a short circuit, as in Fig. 6.27(b). It is evident from Fig. 6.27(b) that

$$i = i_L = \frac{12}{1 + 5} = 2 \text{ A}$$

The voltage v_C is the same as the voltage across the 5-Ω resistor. Hence,

$$v_C = 5i = 10 \text{ V}$$

- (b) The energy in the capacitor is

$$w_C = \frac{1}{2} C v_C^2 = \frac{1}{2} (1) (10^2) = 50 \text{ J}$$

and that in the inductor is

$$w_L = \frac{1}{2} L i_L^2 = \frac{1}{2} (2) (2^2) = 4 \text{ J}$$

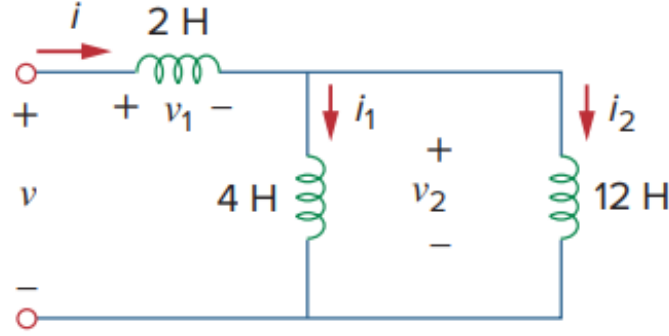
10. **Example. Voltage across each inductor.**

Figure 6.33
For Example 6.12.

Solution:

(a) From $i(t) = 4(2 - e^{-10t})$ mA, $i(0) = 4(2 - 1) = 4$ mA. Since $i = i_1 + i_2$,

$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5 \text{ mA}$$

(b) The equivalent inductance is

$$L_{eq} = 2 + \left(\frac{4 \parallel 12}{1} \right) = 2 + 3 = 5 \text{ H}$$

Thus,

$$v(t) = L_{eq} \frac{di}{dt} = 5(4)(-1)(-10)e^{-10t} = 200e^{-10t} \text{ mV}$$

and

$$v_1(t) = 2 \frac{di}{dt} = 2(-4)(-10)e^{-10t} = 80e^{-10t} \text{ mV}$$

Since $v = v_1 + v_2$,

$$v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$$

(c) The current i_1 is obtained as

$$\begin{aligned} i_1(t) &= \frac{1}{4} \int_0^t v_2 dt + i_1(0) = \frac{120}{4} \int_0^t e^{-10t} dt + 5 \text{ mA} \\ &= -3e^{-10t} \Big|_0^t + 5 \text{ mA} = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t} \text{ mA} \end{aligned}$$

Similarly,

$$\begin{aligned} i_2(t) &= \frac{1}{12} \int_0^t v_2 dt + i_2(0) = \frac{120}{12} \int_0^t e^{-10t} dt - 1 \text{ mA} \\ &= -e^{-10t} \Big|_0^t - 1 = -e^{-10t} + 1 - 1 = -e^{-10t} \text{ mA} \end{aligned}$$

Note that $i_1(t) + i_2(t) = i(t)$.

1.3 Operational amplifier

1.3.1 Introduction

1. An operational amplifier (Op-amp) is an electronic unit that behaves like a **voltage-controlled voltage source**.
2. **Purpose of using op-amp.** An op amp is an **active** circuit element designed to **perform mathematical operations** of addition, subtraction, multiplication, division, differentiation, and integration.

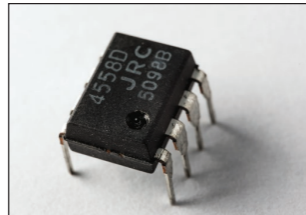


Figure 5.1
A typical operational amplifier.
Mark Dierker/McGraw-Hill Education

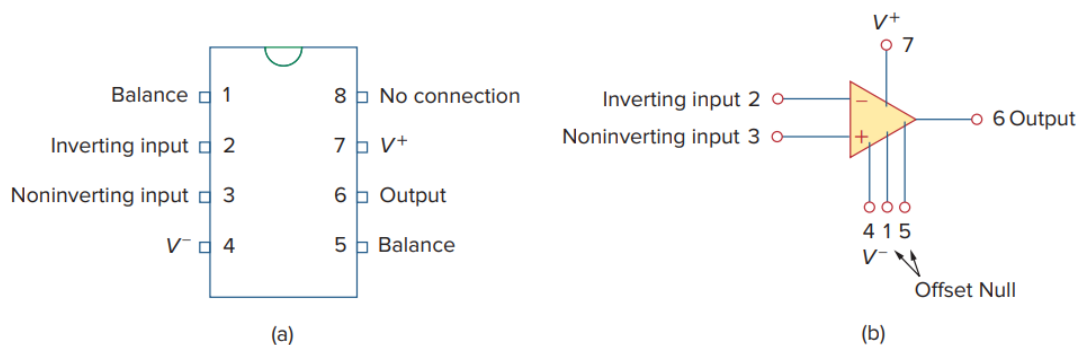


Figure 5.2
A typical op amp: (a) pin configuration, (b) circuit symbol.

3. **Pin-diagram and Powering the op-amp.** Figure 5.1 shows a typical op amp package. A typical one is the eight-pin dual in-line package (or DIP), shown in Fig. 5.2(a). Pin or terminal 8 is unused, and terminals 1 and 5 are of little concern to us. The five important terminals are:

- (a) The inverting input, pin 2.
- (b) The noninverting input, pin 3.
- (c) The output, pin 6.
- (d) The positive power supply V^+ , pin 7.
- (e) The negative power supply V^- , pin 4.

4. **Inverting and non-inverting input.** The op amp has two inputs and one output. The inputs are marked with minus (−) and plus (+) to specify *inverting* and *noninverting* inputs, respectively.

- An input applied to the noninverting terminal will appear with the same polarity at the output.
- an input applied to the inverting terminal will appear inverted at the output.

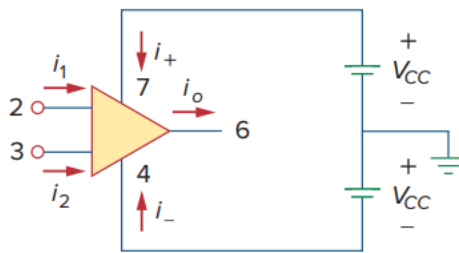


Figure 5.3
Powering the op amp.

5. As an active element, the op amp must be powered by a voltage supply as typically shown in Fig.5.3. Although the power supplies are often ignored in op amp circuit diagrams for the sake of simplicity, the **power supply currents** must not be overlooked. By KCL,

$$i_o = (i_+ + i_-) + (i_1 + i_2)$$

6. **Equivalent circuit model of a nonideal op-amp.** The output section consists

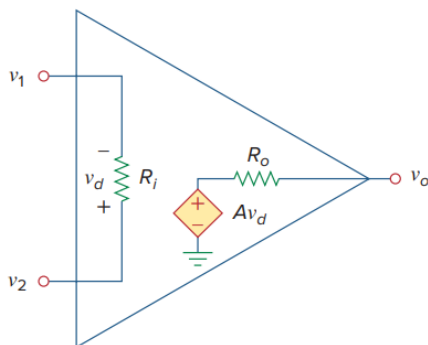


Figure 5.4
The equivalent circuit of the nonideal op amp.

TABLE 5.1

Typical ranges for op amp parameters.

Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to 10^8	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100Ω	0Ω
Supply voltage, V_{CC}	5 to 24 V	

of a voltage-controlled source in series with the output resistance R_o . It is evident from Fig. 5.4 that the input resistance R_i is the Thevenin equivalent resistance seen at the input terminals, while the output resistance R_o is the Thevenin equivalent resistance seen at the output.

The differential input voltage v_d is given by

$$v_d = v_2 - v_1 \quad (5.2)$$

where v_1 is the voltage between the inverting terminal and ground and v_2 is the voltage between the noninverting terminal and ground.

The op amp senses the difference between the two inputs, multiplies it by the gain A , and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

$$v_o = Av_d = A(v_2 - v_1) \quad (5.3)$$

A is called the *open-loop voltage gain* because it is the gain of the op amp without any external feedback from output to input. Table 5.1 shows typical values of voltage gain A , input resistance R_i , output resistance R_o , and supply voltage V_{CC} .

7. Saturation of Op-amp.

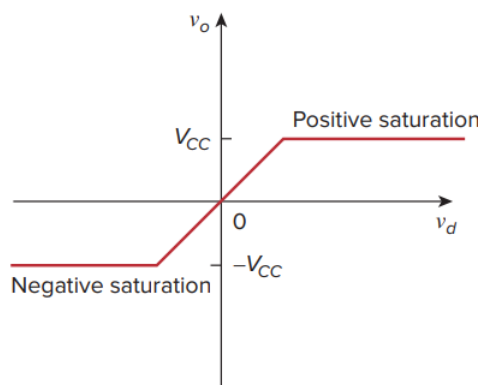


Figure 5.5

Op amp output voltage v_o as a function of the differential input voltage v_d .

A practical limitation of the op amp is that the magnitude of its output voltage cannot exceed $|V_{CC}|$. In other words, the output voltage is dependent on and is limited by the power supply voltage. Figure 5.5 illustrates that the op amp can operate in three modes, depending on the differential input voltage v_d :

- (a) Positive saturation, $v_o = V_{CC}$.
- (b) Linear region, $-V_{CC} \leq v_o = Av_d \leq V_{CC}$.
- (c) Negative saturation, $v_o = -V_{CC}$.

If we attempt to increase v_d beyond the linear range, the op amp becomes saturated and yields $v_o = V_{CC}$ or $v_o = -V_{CC}$. We always assume that our op amps operate in the linear mode. This means that the output voltage is restricted by

$$-V_{CC} \leq v_o \leq V_{CC}$$

8. Negative feedback and closed loop gain.

The concept of feedback is crucial to our understanding of op amp circuits. A negative feedback is achieved when the output is fed back to the inverting terminal of the op amp. As the following example shows, when there is a feedback path from output to input, the ratio of the output voltage to the input voltage is called the **closed-loop gain**. As a result of the negative feedback, it can be shown that the closed-loop gain is almost **insensitive to the open-loop gain** A of the op amp. For this reason, op amps are used in circuits with feedback paths.

Example 1. Closed loop gain of an non-ideal op-amp.

A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2 \text{ V}$.

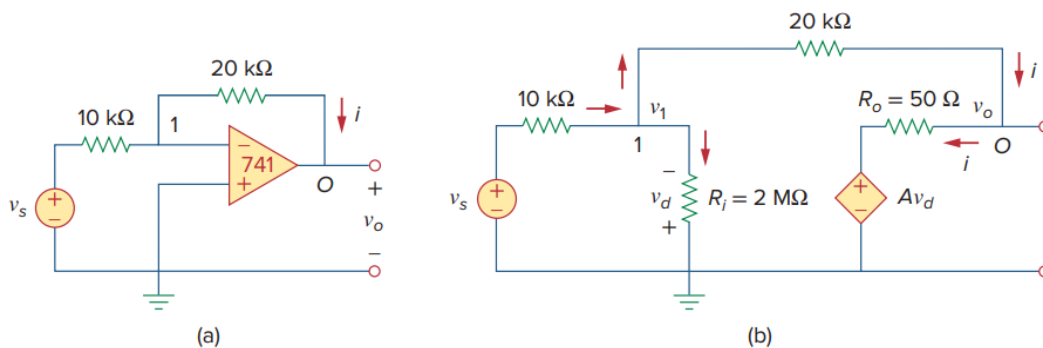


Figure 5.6

For Example 5.1: (a) original circuit, (b) the equivalent circuit.

Solution. Using the op amp model in Fig. 5.4, we obtain the equivalent circuit of Fig. 5.6(a) as shown in Fig. 5.6(b). We now solve the circuit in Fig. 5.6(b) by using nodal analysis. At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

Multiplying through by 2000×10^3 , we obtain

$$200v_s = 301v_1 - 100v_o$$

or

$$2v_s \simeq 3v_1 - v_o \quad \Rightarrow \quad v_1 = \frac{2v_s + v_o}{3} \quad (5.1.1)$$

At node O ,

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

But $v_d = -v_1$ and $A = 200,000$. Then

$$v_1 - v_o = 400(v_o + 200,000v_1) \quad (5.1.2)$$

Substituting v_1 from Eq. (5.1.1) into Eq. (5.1.2) gives

$$0 \simeq 26,667,067v_o + 53,333,333v_s \Rightarrow \frac{v_o}{v_s} = -1.9999699$$

This is closed-loop gain, because the $20\text{ k}\Omega$ feedback resistor closes the loop between the output and input terminals. When $v_s = 2\text{ V}$, $v_o = -3.999938\text{ V}$. From Eq. (5.1.1), we obtain $v_1 = 20.066667\text{ }\mu\text{V}$. Thus,

$$i = \frac{v_1 - v_o}{20 \times 10^3} = 0.19999\text{ mA}$$

It is evident that working with a nonideal op amp is tedious, as we are dealing with very large numbers.

Example 2. Closed loop gain of an non-ideal op-amp.

If the same 741 op amp in Example 5.1 is used in the circuit of Fig. 5.7, calculate the closed-loop gain v_o/v_s . Find i_o when $v_s = 1\text{ V}$.

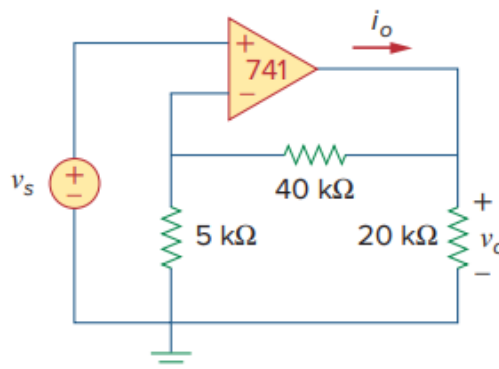


Figure 5.7
For Practice Prob. 5.1.

Answer: 9.00041, $657\text{ }\mu\text{A}$

1.3.2 Ideal Op-amp

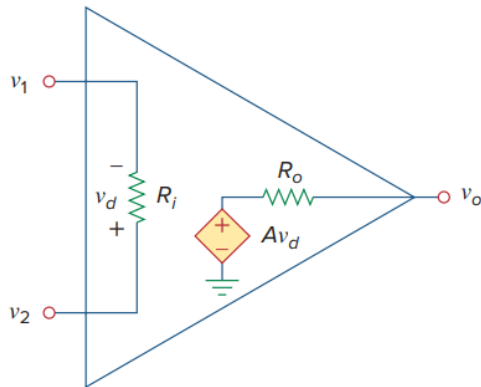


Figure 5.4

The equivalent circuit of the nonideal op amp.

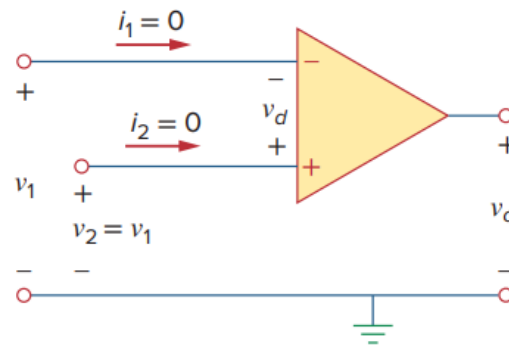


Figure 5.8

Ideal op amp model.

1. The following **ideal** op amp approximation greatly facilitates circuit analysis involving op-amps.
 - (a) Infinite open-loop gain, $A \simeq \infty$.
 - (b) Infinite input resistance, $R_i \simeq \infty$.
 - (c) Zero output resistance, $R_o \simeq 0$.

2. This approximation yields two important properties of an ideal op-amp,

- Infinite input resistance ($R_i \simeq \infty$) implies $i_1 = i_2 = 0$. The currents into both input terminals are zero, since an open circuit exists there and current cannot enter the op amp. But the output current is not necessarily zero since

$$i_o = (i_1 + i_2) + (i_+ + i_-) = 0 + 0 + (i_+ + i_-) = (i_+ + i_-)$$

- The voltage across the input terminals is equal to zero; i.e., $v_d = v_1 - v_2 = 0$ or

$$v_1 = v_2$$

3. **Example. Ideal-op amp approximation.**

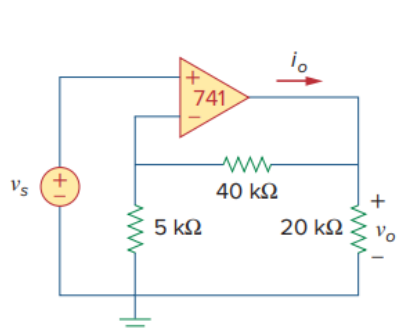


Figure 5.7

For Practice Prob. 5.1.

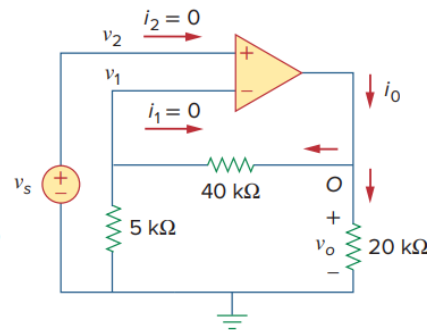


Figure 5.9

For Example 5.2.

Using ideal op-amp approximation, calculate the closed-loop gain v_o/v_s . Find i_o when $v_s = 1$ V.

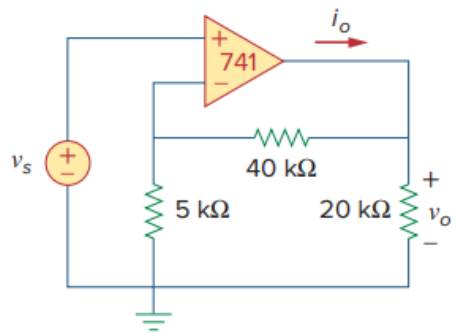


Figure 5.7
For Practice Prob. 5.1.

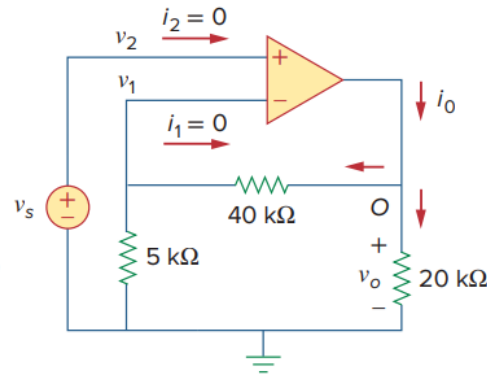


Figure 5.9
For Example 5.2.

Solution: We may replace the op amp in Fig. 5.7 by its equivalent model in Fig. 5.9 as we did in Example 5.1. But we do not really need to do this. We just need to keep Eqs. (5.5) and (5.7) in mind as we analyze the circuit in Fig. 5.7. Thus, the Fig. 5.7 circuit is presented as in Fig. 5.9. Notice that

$$v_2 = v_s \quad (5.2.1)$$

Since $i_1 = 0$, the 40- and 5-k Ω resistors are in series; the same current flows through them. v_1 is the voltage across the 5-k Ω resistor. Hence, using the voltage division principle,

$$v_1 = \frac{5}{5 + 40} v_o = \frac{v_o}{9} \quad (5.2.2)$$

According to Eq. (5.7),

$$v_2 = v_1 \quad (5.2.3)$$

Substituting Eqs. (5.2.1) and (5.2.2) into Eq. (5.2.3) yields the closed-loop gain,

$$v_s = \frac{v_o}{9} \Rightarrow \frac{v_o}{v_s} = 9 \quad (5.2.4)$$

which is very close to the value of 9.00041 obtained with the nonideal model in Fig. 5.7. This shows that negligibly small error results from assuming ideal op amp characteristics.

At node O,

$$i_o = \frac{v_o}{40 + 5} + \frac{v_o}{20} \text{ mA} \quad (5.2.5)$$

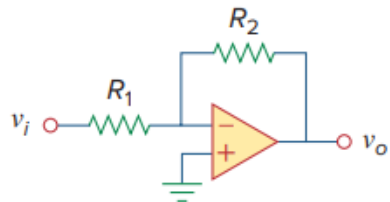
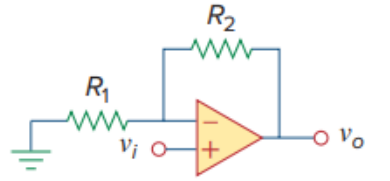
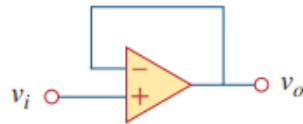
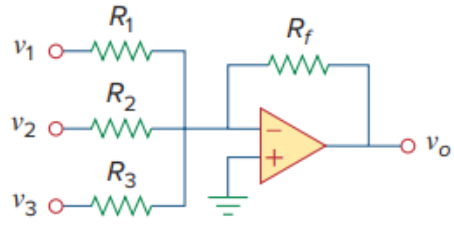
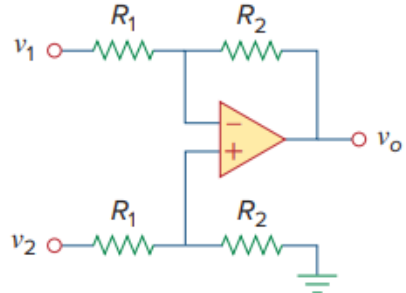
From Eq. (5.2.4), when $v_s = 1 \text{ V}$, $v_o = 9 \text{ V}$. Substituting for $v_o = 9 \text{ V}$ in Eq. (5.2.5) produces

$$i_o = 0.2 + 0.45 = 0.65 \text{ mA}$$

This, again, is close to the value of 0.657 mA obtained in Fig. 5.7 with the nonideal model.

1.4 Practical Op-amp circuits

1.4.1 Basic mathematical operations with op-amp

Op amp circuit	Name/output-input relationship
	Inverting amplifier $v_o = -\frac{R_2}{R_1}v_i$
	Noninverting amplifier $v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$
	Voltage follower $v_o = v_i$
	Summer $v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$
	Difference amplifier $v_o = \frac{R_2}{R_1}(v_2 - v_1)$

Inverting amplifier

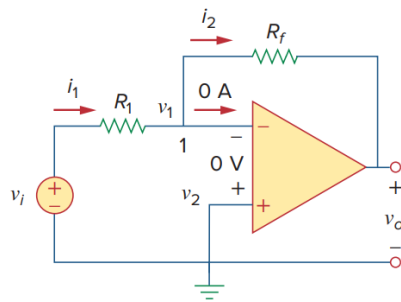


Figure 5.10
The inverting amplifier.

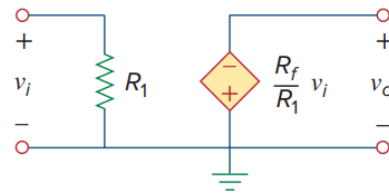


Figure 5.11
An equivalent circuit for the inverter in Fig. 5.10.

1. **Obtaining closed loop gain A_v using nodal analysis.** In Fig. 5.10, applying KCL at node 1,

$$i_1 = i_2 \Rightarrow \frac{v_i - v_1}{R_1} = \frac{v_1 - v_o}{R_f} \quad (5.8)$$

But $v_1 = v_2 = 0$ for an ideal op amp, since the noninverting terminal is grounded. Hence,

$$\frac{v_i}{R_1} = -\frac{v_o}{R_f}$$

or

$$v_o = -\frac{R_f}{R_1} v_i \quad (5.9)$$

The voltage gain is $A_v = \frac{v_o}{v_i} = -\frac{R_f}{R_1}$. The designation of the circuit in Fig. 5.10 as an *inverter* arises from the negative sign.

2. An **inverting amplifier** reverses the polarity of the input signal while amplifying it. Note that both the input signal and the feedback are applied at the **inverting terminal** of the op amp.
3. **Example.** Determine v_o in the op amp circuit shown in Fig. 5.14. (**Ans.** $v_o = -6 \text{ V}$)

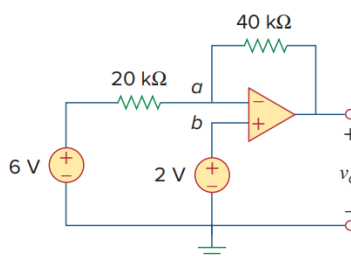


Figure 5.14
For Example 5.4.

4. **Example. Current-to-Voltage converters.** By applying nodal analysis, we can

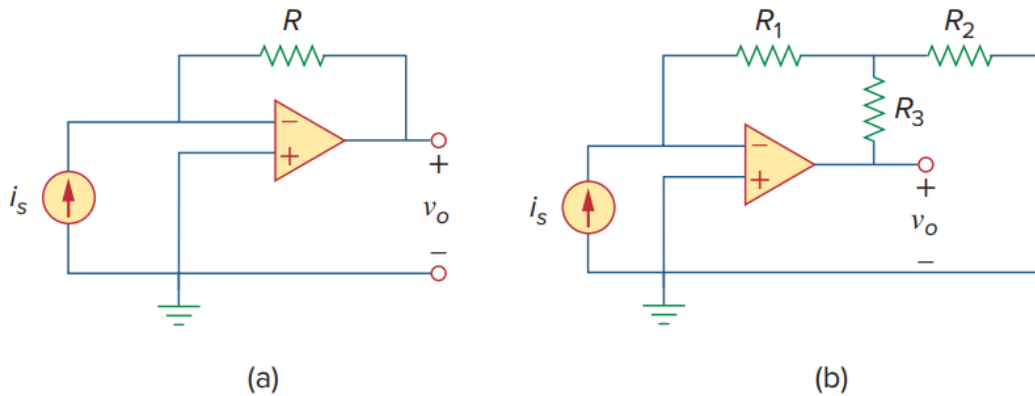


Figure 5.15

For Practice Prob. 5.4.

show that

(a) For the converter in Fig. 5.15(a),

$$\frac{v_o}{i_s} = -R$$

(b) For the converter in Fig. 5.15(b),

$$\frac{v_o}{i_s} = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)$$

Non-inverting amplifier

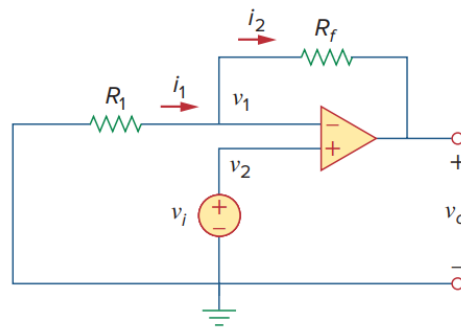


Figure 5.16
The noninverting amplifier.

1. **Obtaining the closed loop gain.** Application of KCL at the inverting terminal gives

$$i_1 = i_2 \Rightarrow \frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

But $v_1 = v_2 = v_i$. Then

$$\frac{-v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

or

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i \quad (5.11)$$

The voltage gain is $A_v = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_1}$, which does not have a negative sign. Thus, the output has the same polarity as the input.

2. A **noninverting amplifier** is an op amp circuit designed to provide a positive voltage gain.

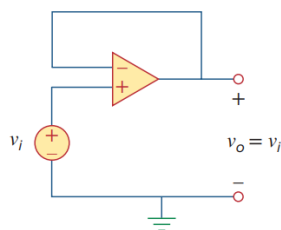


Figure 5.17
The voltage follower.

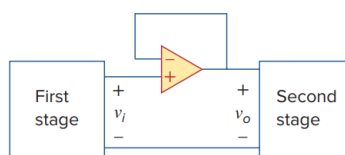


Figure 5.18
A voltage follower used to isolate two cascaded stages of a circuit.

3. In the limit of infinite input resistance ($R_1 \rightarrow \infty$) and zero feedback resistance ($R_f \rightarrow 0$), the noninverting amplifier becomes a **voltage follower**, because the output follows the input

$$v_o = v_i$$

Such a circuit has a very high input impedance and is therefore useful as an intermediate-stage (or buffer) amplifier to isolate one circuit from another, as portrayed in Fig. 5.18. The voltage follower minimizes interaction between the two stages and eliminates interstage loading.

4. **Example.** For the op amp circuit in Fig. 5.19, calculate the output voltage v_o .

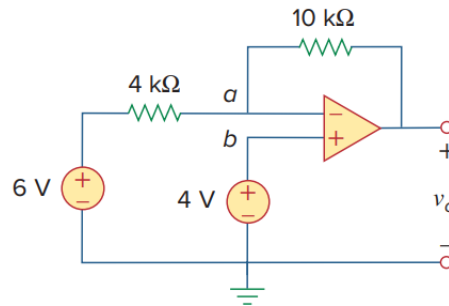


Figure 5.19
For Example 5.5.

Solution 1. Superposition. Using superposition, we let

$$v_o = v_{o1} + v_{o2}$$

where v_{o1} is due to the 6-V voltage source, and v_{o2} is due to the 4-V input.

To get v_{o1} , we set the 4-V source equal to zero. Under this condition, the circuit becomes an inverter. Hence Eq. (5.9) gives

$$v_{o1} = \frac{-10}{4}(6) = -15 \text{ V}$$

To get v_{o2} , we set the 6-V source equal to zero. The circuit becomes a noninverting amplifier so that

$$v_{o2} = \left(1 + \frac{10}{4}\right) 4 = 14 \text{ V}$$

Thus,

$$v_o = v_{o1} + v_{o2} = -15 + 14 = -1 \text{ V}$$

Solution 2. Nodal analysis

Applying KCL at node a ,

$$\frac{6 - v_a}{4} = \frac{v_a - v_o}{10}$$

But $v_a = v_b = 4$, and so

$$\frac{6 - 4}{4} = \frac{4 - v_o}{10} \Rightarrow 5 = 4 - v_o$$

or

$$v_o = -1 \text{ V, as before.}$$

Summing amplifier

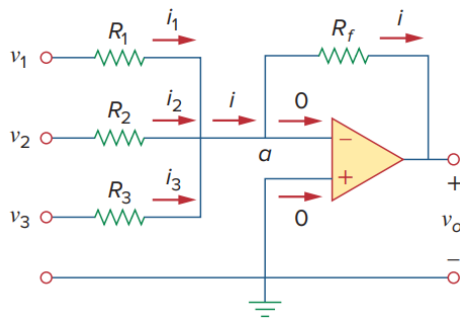


Figure 5.21
The summing amplifier.

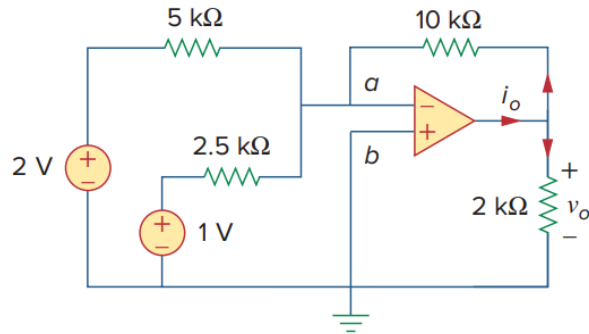


Figure 5.22
For Example 5.6.

1. A **summing amplifier** is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs.
2. A summing amplifier with N input can be understood as a superposition of N inverting amplifiers.
3. In the case of three inputs, applying KCL at node a gives

$$i = i_1 + i_2 + i_3 \quad (5.13)$$

But

$$i_1 = \frac{v_1 - v_a}{R_1}, \quad i_2 = \frac{v_2 - v_a}{R_2}, \quad i_3 = \frac{v_3 - v_a}{R_3}, \quad i = \frac{v_a - v_o}{R_f} \quad (5.14)$$

We note that $v_a = 0$ and substitute Eq. (5.14) into Eq. (5.13). We get

$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

4. **Example.** Calculate v_o and i_o in the op amp circuit in Fig. 5.22.

Ans. $v_o = -8 \text{ V}$, $i_o = -4.8 \text{ mA}$

5. Application of a summer amplifier: Digital-to-Analog converter (DAC).

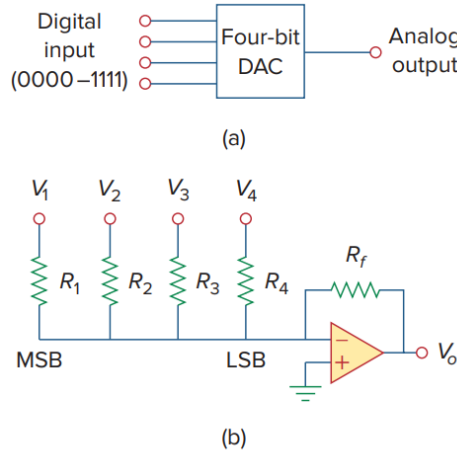


Figure 5.36
Four-bit DAC: (a) block diagram,
(b) binary weighted ladder type.

The digital-to-analog converter (DAC) transforms digital signals into analog form. A typical example of a four-bit DAC is illustrated in Fig. 5.36(a). The four-bit DAC can be realized in many ways. A simple realization is the *binary weighted ladder*, shown in Fig. 5.36(b). The bits are weights according to the magnitude of their place value, by descending value of R_f/R_n so that each lesser bit has half the weight of the next higher. This is obviously an inverting summing amplifier. The output is related to the inputs as shown in Eq. (5.15). Thus,

$$-V_o = \frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \frac{R_f}{R_4}V_4 \quad (5.23)$$

Input V_1 is called the *most significant bit* (MSB), while input V_4 is the *least significant bit* (LSB). Each of the four binary inputs $V_1, \dots, V_4$ can assume only two voltage levels: 0 or 1 V. By using the proper input and feedback resistor values, the DAC provides a single output that is proportional to the inputs.

Example. In the op amp circuit of Fig. 5.36(b), let $R_f = 10 \text{ k}\Omega$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $R_3 = 40 \text{ k}\Omega$, and $R_4 = 80 \text{ k}\Omega$. Obtain the analog output for binary inputs [0000], [0001], [0010], ..., [1111].

Solution:

Substituting the given values of the input and feedback resistors in Eq. (5.23) gives

$$-V_o = \frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \frac{R_f}{R_4}V_4 = V_1 + 0.5V_2 + 0.25V_3 + 0.125V_4$$

Using this equation, a digital input $[V_1V_2V_3V_4] = [0000]$ produces an analog output of $-V_o = 0 \text{ V}$;

$[V_1V_2V_3V_4] = [0001]$ gives $-V_o = 0.125 \text{ V}$

Similarly,

$$[V_1 V_2 V_3 V_4] = [0010] \Rightarrow -V_o = 0.25 \text{ V}$$

$$[V_1 V_2 V_3 V_4] = [0011] \Rightarrow -V_o = 0.25 + 0.125 = 0.375 \text{ V}$$

$$[V_1 V_2 V_3 V_4] = [0100] \Rightarrow -V_o = 0.5 \text{ V}$$

$$\vdots$$

$$[V_1 V_2 V_3 V_4] = [1111] \Rightarrow -V_o = 1 + 0.5 + 0.25 + 0.125 = 1.875 \text{ V}$$

TABLE 5.2

Input and output values of the four-bit DAC.

Binary input [$V_1 V_2 V_3 V_4$]	Decimal value	Output $-V_o$
0000	0	0
0001	1	0.125
0010	2	0.25
0011	3	0.375
0100	4	0.5
0101	5	0.625
0110	6	0.75
0111	7	0.875
1000	8	1.0
1001	9	1.125
1010	10	1.25
1011	11	1.375
1100	12	1.5
1101	13	1.625
1110	14	1.75
1111	15	1.875

Table 5.2 summarizes the result of the digital-to-analog conversion. Note that we have assumed that each bit has a value of 0.125 V. Thus, in this system, we cannot represent a voltage between 1.000 and 1.125, for example. This lack of resolution is a major limitation of digital-to-analog conversions. For greater accuracy, a word representation with a greater number of bits is required. Even then a digital representation of an analog voltage is never exact. In spite of this inexact representation, digital representation has been used to accomplish remarkable things such as audio CDs and digital photography.

1.4.2 Difference Amplifier

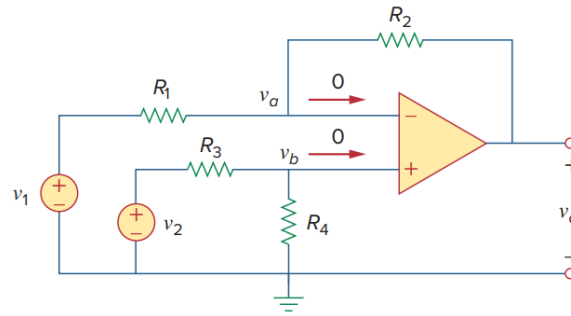


Figure 5.24
Difference amplifier.

1. A **difference amplifier** is a device that amplifies the difference between two inputs but **rejects** any signals common to the two inputs (common mode signal rejection.)
2. **Obtaining closed loop gain using nodal analysis.** Applying KCL to node b ,

$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$$

$$v_b = \frac{R_4}{R_3 + R_4} v_2 \quad (5.17)$$

But $v_a = v_b$. Substituting Eq. (5.17) into Eq. (5.16) yields

$$v_o = \left(\frac{R_2}{R_1} + 1 \right) \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$$

or

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1 \quad (5.18)$$

Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that $v_o = 0$ when $v_1 = v_2$. This property exists when

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (5.19)$$

Thus, when the op amp circuit is a difference amplifier, Eq. (5.18) becomes

$$v_o = \frac{R_2}{R_1} (v_2 - v_1) \quad (5.20)$$

If $R_2 = R_1$ and $R_3 = R_4$, the difference amplifier becomes a *subtractor*, with the output

$$v_o = v_2 - v_1$$

3. Design example.

Design an op amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$.

Solution:

The circuit requires that

$$v_o = 3v_2 - 5v_1 \quad (5.7.1)$$

This circuit can be realized in two ways.

Design 1

If we desire to use only one op amp, we can use the op amp circuit of Fig. 5.24. Comparing Eq. (5.7.1) with Eq. (5.18), we see

$$\frac{R_2}{R_1} = 5 \Rightarrow R_2 = 5R_1 \quad (5.7.2)$$

Also,

$$5 \left(\frac{1 + R_1/R_2}{1 + R_3/R_4} \right) = 3 \Rightarrow \frac{6/5}{1 + R_3/R_4} = \frac{3}{5}$$

or

$$2 = 1 + \frac{R_3}{R_4} \Rightarrow R_3 = R_4 \quad (5.7.3)$$

If we choose $R_1 = 10 \text{ k}\Omega$ and $R_3 = 20 \text{ k}\Omega$, then $R_2 = 50 \text{ k}\Omega$ and $R_4 = 20 \text{ k}\Omega$.

Design 2

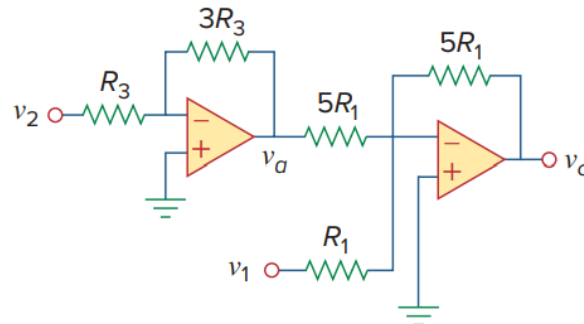


Figure 5.25
For Example 5.7.

If we desire to use more than one op amp, we may cascade an inverting amplifier and a two-input inverting summer, as shown in Fig. 5.25. For the summer,

$$v_o = -v_a - 5v_1 \quad (5.7.4)$$

and for the inverter,

$$v_a = -3v_2 \quad (5.7.5)$$

Combining Eqs. (5.7.4) and (5.7.5) gives

$$v_o = 3v_2 - 5v_1$$

which is the desired result. In Fig. 5.25, we may select $R_1 = 10 \text{ k}\Omega$ and $R_3 = 20 \text{ k}\Omega$ or $R_1 = R_3 = 10 \text{ k}\Omega$.

4. Application of a difference amplifier using nodal analysis.

One of the most useful and versatile op amp circuits for precision measurement and process control is the *instrumentation amplifier* (IA), so called because of its widespread use in measurement systems. Typical applications of IAs include isolation amplifiers, thermocouple amplifiers, and data acquisition systems.

The instrumentation amplifier is an extension of the difference amplifier in that it amplifies the difference between its input signals. As shown in Fig. 5.26, an instrumentation amplifier typically consists of three op amps and seven resistors.

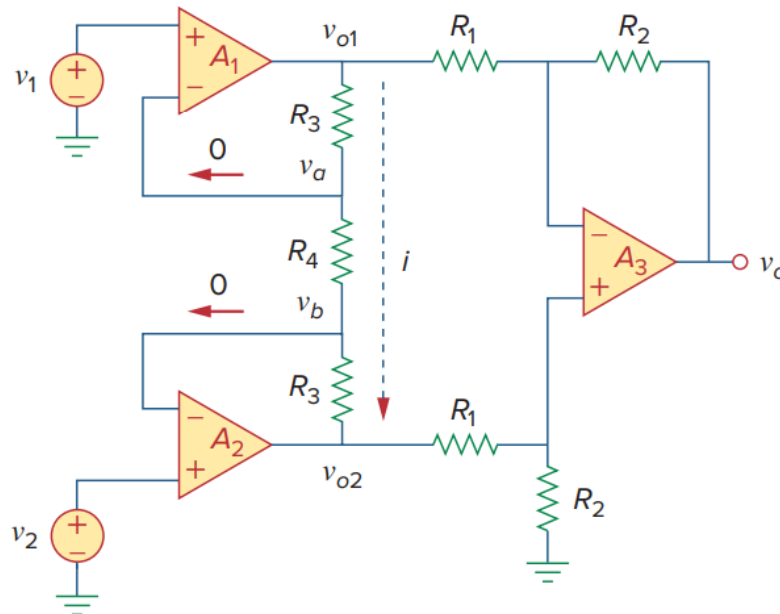


Figure 5.26

Instrumentation amplifier; for Example 5.8.

Show that

$$v_o = \frac{R_2}{R_1} \left(1 + \frac{2R_3}{R_4} \right) (v_2 - v_1)$$

Solution:

We recognize that the amplifier A_3 in Fig. 5.26 is a difference amplifier. Thus,

$$v_o = \frac{R_2}{R_1} (v_{o2} - v_{o1}) \quad (5.8.1)$$

Since the op amps A_1 and A_2 draw no current, current i flows through the three resistors as though they were in series. Hence,

$$v_{o1} - v_{o2} = i(R_3 + R_4 + R_3) = i(2R_3 + R_4) \quad (5.8.2)$$

But

$$i = \frac{v_a - v_b}{R_4}$$

and $v_a = v_1$, $v_b = v_2$. Therefore,

$$i = \frac{v_1 - v_2}{R_4} \quad (5.8.3)$$

Inserting Eqs. (5.8.2) and (5.8.3) into Eq. (5.8.1) gives

$$v_o = \frac{R_2}{R_1} \left(1 + \frac{2R_3}{R_4} \right) (v_2 - v_1)$$

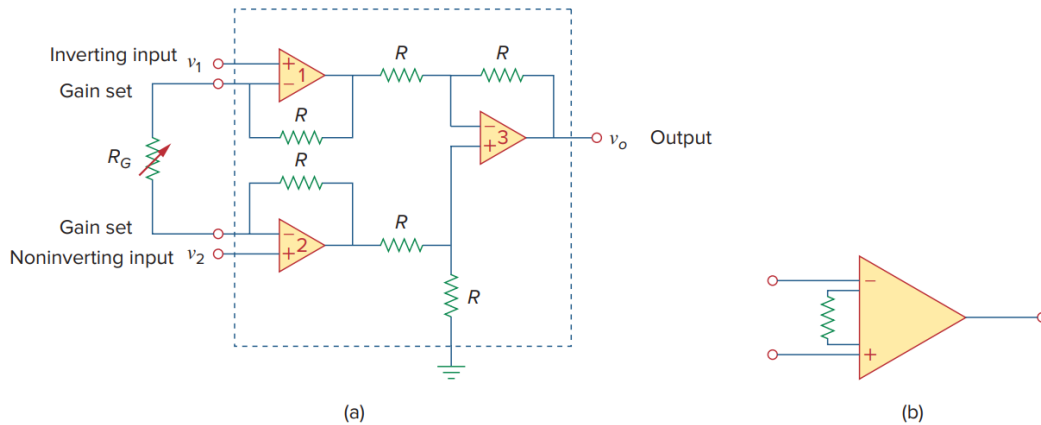


Figure 5.38

(a) The instrumentation amplifier with an external resistance to adjust the gain, (b) schematic diagram.

For convenience, the amplifier is shown again in Fig. 5.38(a), where the resistors are made equal except for the external gain-setting resistor R_G , connected between the gain set terminals. Figure 5.38(b) shows its schematic symbol. In the previous derivation of v_o , setting $R_1 = R_2 = R_3 = R$ and $R_4 = R_G$ gives

$$v_o = A_v(v_2 - v_1) \quad (5.24)$$

where the voltage gain is

$$A_v = 1 + \frac{2R}{R_G} \quad (5.25)$$

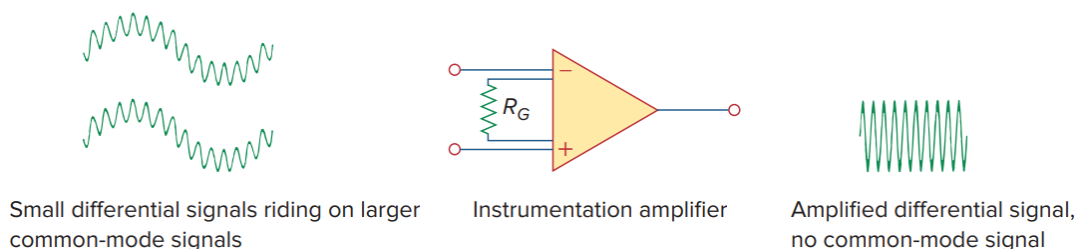


Figure 5.39

The IA rejects common voltages but amplifies small signal voltages.

As shown in Fig. 5.39, the instrumentation amplifier amplifies small differential signal voltages superimposed on larger common-mode voltages. Since the common-mode voltages are equal, they cancel each other.

The IA has three major characteristics:

- (a) The voltage gain is adjusted by *one* external resistor R_G .
- (b) The input impedance of both inputs is very high and does not vary as the gain is adjusted.
- (c) The output v_o depends on the difference between the inputs v_1 and v_2 , not on the voltage common to them (common-mode voltage).

Due to the widespread use of IAs, manufacturers have developed these amplifiers on single-package units. A typical example is the LH0036, developed by National Semiconductor. The gain can be varied from 1 to 1,000 by an external resistor whose value may vary from 100 to 10 k Ω .

1.4.3 Cascading op-amp circuits

1. It is often necessary in practical applications to connect op amp circuits in cascade (i.e., head to tail) to achieve a large overall gain. In general, two circuits are cascaded when they are connected in tandem, one behind another in a single file.

A **cascade connection** is a head-to-tail arrangement of two or more op amp circuits such that the output of one is the input of the next.

2. When op amp circuits are cascaded, each circuit in the string is called a stage; the original input signal is increased by the gain of the individual stage.
3. Op amp circuits have the advantage that they can be cascaded without changing their input-output relationships. This is due to the fact that each (ideal) op amp circuit has infinite input resistance and zero output resistance.

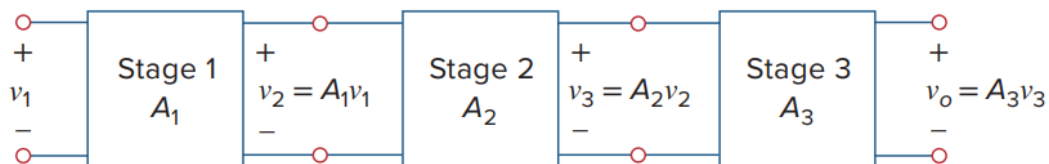


Figure 5.28

A three-stage cascaded connection.

4. Figure 5.28 displays a block diagram representation of three op amp circuits in cascade. Since the output of one stage is the input to the next stage, the overall gain of the cascade connection is the product of the gains of the individual op amp circuits, or

$$A = A_1 A_2 A_3$$

5. **Caution.** Although the cascade connection does not affect the op amp input-output relationships, care must be exercised in the design of an actual op amp circuit to ensure that the load due to the next stage in the cascade does not saturate the op amp.

6. **Example 1.** Find v_o and i_o in the circuit in Fig. 5.29.

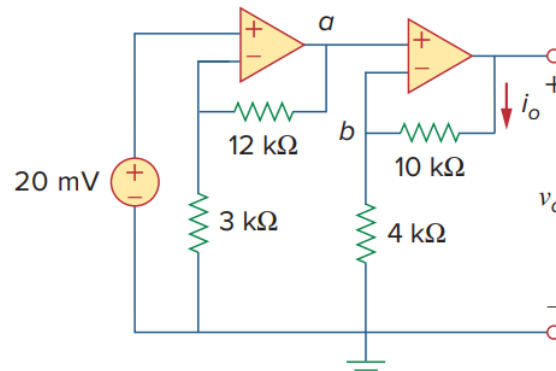


Figure 5.29
For Example 5.9.

Solution:

This circuit consists of two noninverting amplifiers cascaded. At the output of the first op amp,

$$v_a = \left(1 + \frac{12}{3}\right) (20) = 100 \text{ mV}$$

At the output of the second op amp,

$$v_o = \left(1 + \frac{10}{4}\right) v_a = (1 + 2.5)100 = 350 \text{ mV}$$

The required current i_o is the current through the 10-k Ω resistor.

$$i_o = \frac{v_o - v_b}{10} \text{ mA}$$

But $v_b = v_a = 100 \text{ mV}$. Hence,

$$i_o = \frac{(350 - 100) \times 10^{-3}}{10 \times 10^3} = 25 \text{ } \mu\text{A}$$

7. **Example 2.** If $v_1 = 1V$ and $v_2 = 2V$, find v_o in the op amp circuit of Fig. 5.31.

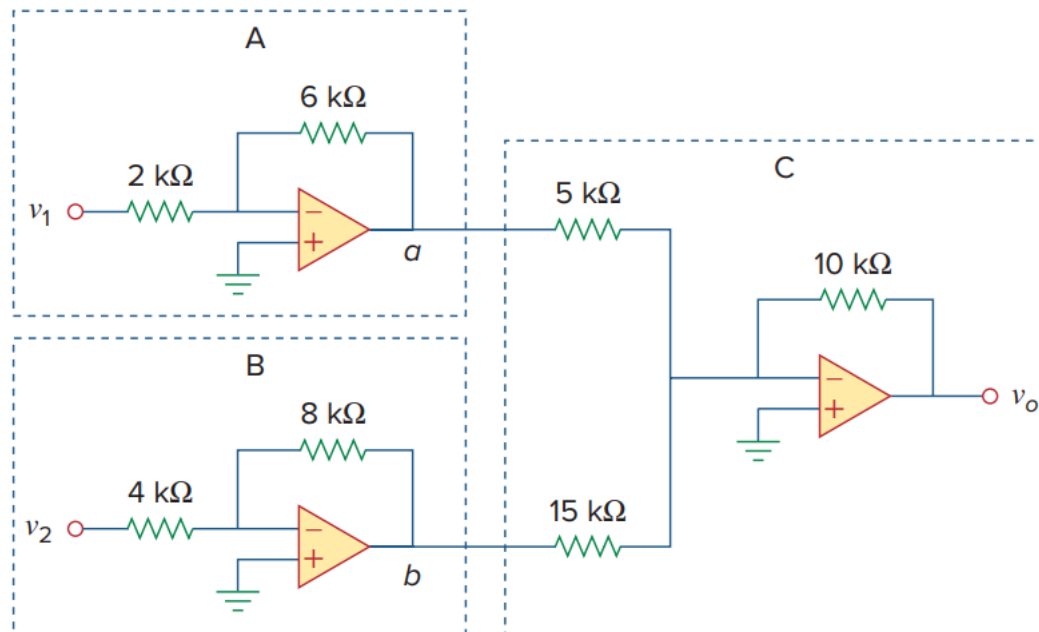


Figure 5.31
For Example 5.10.

Ans. 8.667 V

1.4.4 Op-amp circuits with capacitors

1. **Motivation.** Circuit elements such as resistors and capacitors are commercially available in either discrete form or integrated-circuit (IC) form. Unlike capacitors and resistors, inductors with appreciable inductance are difficult to produce on IC substrates. Therefore, inductors (coils) usually come in discrete form and tend to be more bulky and expensive. For this reason, inductors are not as versatile as capacitors and resistors, and they are more limited in applications. However, there are several applications in which inductors have no practical substitute. They are routinely used in relays, delays, sensing devices, pick-up heads, telephone circuits, radio and TV receivers, power supplies, electric motors, microphones, and loudspeakers, to mention a few.
2. Capacitors and inductors possess the following three special properties that make them very useful in electric circuits:

The capacity to store energy makes them useful as temporary voltage or current sources. Thus, they can be used for generating a large amount of current or voltage for a short period of time.

Capacitors oppose any abrupt change in voltage, while inductors oppose any abrupt change in current. This property makes inductors useful for spark or arc suppression and for converting pulsating dc voltage into relatively smooth dc voltage.

Capacitors and inductors are frequency sensitive. This property makes them useful for frequency discrimination.

3. Important op amp circuits that use energy-storage elements include integrators and differentiators. These op amp circuits often involve resistors and capacitors; inductors (coils) tend to be more bulky and expensive.

Integrator

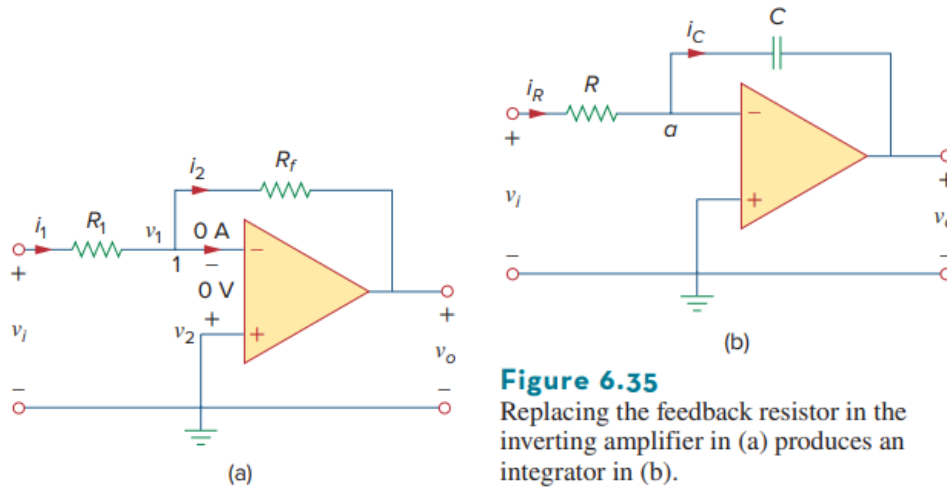


Figure 6.35
Replacing the feedback resistor in the inverting amplifier in (a) produces an integrator in (b).

1. An integrator is an op amp circuit whose output is proportional to the integral of the input signal.
2. If the feedback resistor R_f in the familiar inverting amplifier of Fig. 6.35(a) is replaced by a capacitor, we obtain an ideal integrator, as shown in Fig. 6.35(b). It is interesting that we can obtain a mathematical representation of integration this way. At node a in Fig. 6.35(b),

$$i_R = i_C \quad (6.32)$$

But

$$i_R = \frac{v_i}{R}, \quad i_C = -C \frac{dv_o}{dt}$$

Substituting these in Eq. (6.32), we obtain

$$\frac{v_i}{R} = -C \frac{dv_o}{dt} \quad (6.33a)$$

$$dv_o = -\frac{1}{RC} v_i dt \quad (6.33b)$$

Integrating both sides gives

$$v_o(t) - v_o(0) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau \quad (6.34)$$

To ensure that $v_o(0) = 0$, it is always necessary to discharge the integrator's capacitor prior to the application of a signal. Assuming $v_o(0) = 0$,

$$v_o = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau \quad (6.35)$$

which shows that the circuit in Fig. 6.35(b) provides an output voltage proportional to the integral of the input. In practice, the op amp integrator requires a feedback resistor to reduce dc gain and prevent saturation. Care must be taken that the op amp operates within the linear range so that it does not saturate.

3. **Example.** If $v_1 = 10 \cos 2t$ mV and $v_2 = 0.5t$ mV, find v_o in the op amp circuit in

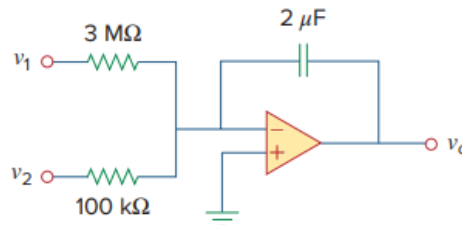


Figure 6.36
For Example 6.13.

Fig. 6.36. Assume that the voltage across the capacitor is initially zero.

Solution:

This is a summing integrator, and

$$\begin{aligned}
 v_o &= -\frac{1}{R_1 C} \int v_1 dt - \frac{1}{R_2 C} \int v_2 dt \\
 &= -\frac{1}{3 \times 10^6 \times 2 \times 10^{-6}} \int_0^t 10 \cos(2\tau) d\tau - \frac{1}{100 \times 10^3 \times 2 \times 10^{-6}} \int_0^t 0.5\tau d\tau \\
 &= -\frac{1}{6} \cdot \frac{10}{2} \sin 2t - \frac{1}{0.2} \cdot \frac{0.5t^2}{2} = -0.833 \sin 2t - 1.25t^2 \text{ mV}
 \end{aligned}$$

Differentiator

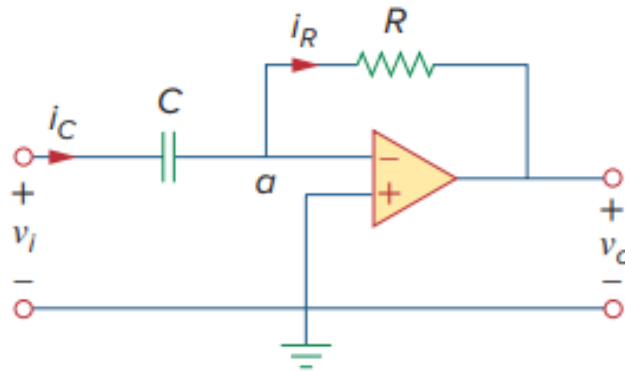


Figure 6.37
An op amp differentiator.

1. In Fig. 6.35(a), if the input resistor is replaced by a capacitor, the resulting circuit is a differentiator, shown in Fig. 6.37. Applying KCL at node a ,

$$i_R = i_C \quad (6.36)$$

But

$$i_R = -\frac{v_o}{R}, \quad i_C = C \frac{dv_i}{dt}$$

Substituting these in Eq. (6.36) yields

$$v_o = -RC \frac{dv_i}{dt} \quad (6.37)$$

showing that the output is the derivative of the input.

2. Differentiator circuits are electronically unstable because any electrical noise within the circuit is exaggerated by the differentiator. For this reason, the differentiator circuit in Fig. 6.37 is not as useful and popular as the integrator. It is seldom used in practice.

3. Example.

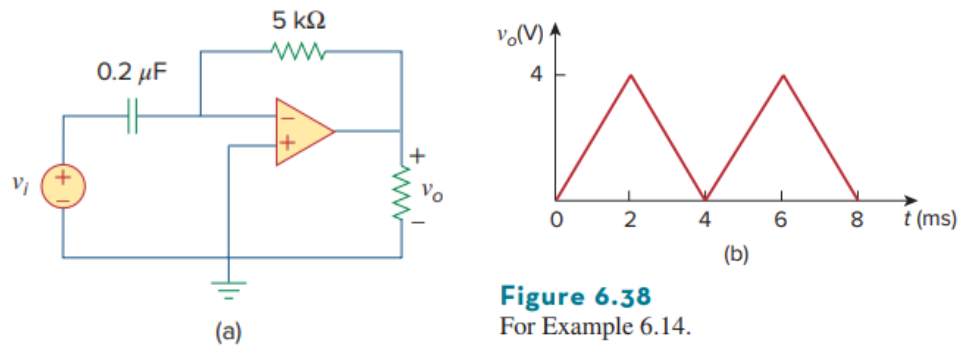


Figure 6.38
For Example 6.14.

Sketch the output voltage for the circuit in Fig. 6.38(a), given the input voltage in Fig. 6.38(b). Take $v_o = 0$ at $t = 0$.

Solution:

This is a differentiator with

$$RC = 5 \times 10^3 \times 0.2 \times 10^{-6} = 10^{-3} \text{ s}$$

For $0 < t < 4$ ms, we can express the input voltage in Fig. 6.38(b) as

$$v_i = \begin{cases} 2000t & 0 < t < 2 \text{ ms} \\ 8 - 2000t & 2 < t < 4 \text{ ms} \end{cases}$$

This is repeated for $4 < t < 8$ ms. Using Eq. (6.37), the output is obtained as

$$v_o = -RC \frac{dv_i}{dt} = \begin{cases} -2 \text{ V} & 0 < t < 2 \text{ ms} \\ 2 \text{ V} & 2 < t < 4 \text{ ms} \end{cases}$$

Thus, the output is as sketched in Fig. 6.39.

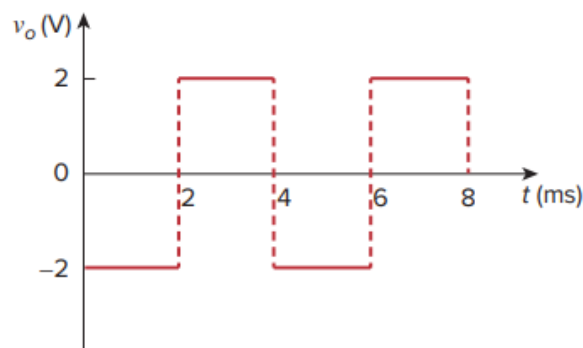


Figure 6.39
Output of the circuit in Fig. 6.38(a).